

00 · Foundations — the words the cookbook assumes

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The notebook that comes **before** everything else. It builds the core vocabulary — potential outcomes, counterfactuals, confounding, randomization, identification, the do-operator, DAGs, ATE vs CATE — from nothing, with one marketing example, and ends with a worked **estimator ladder**: naive $\rightarrow$ regression-adjust $\rightarrow$ IPW $\rightarrow$ doubly-robust — the naive rung fails on purpose, and each adjustment rung then recovers a known effect from confounded data.

No new theory here — it mirrors `docs/causal_inference_primer.pdf`. Skim it once and every later notebook reads as a *named tool* rather than new vocabulary.

1.1 1 · Treatment, unit, outcome — and the counterfactual

A **treatment** is any action whose effect we want (send a discount, show an ad, change a price). The **unit** receives it (customer, store, region); the **outcome** is what we measure after.

For binary $T \in \{0, 1\}$ each unit has **two potential outcomes**: $Y(1)$ if treated, $Y(0)$ if not. We only observe **one** — $Y = TY(1) + (1 - T)Y(0)$ — so the individual effect $Y(1) - Y(0)$ is never seen. This **fundamental problem of causal inference** is a *missing-data* problem; everything downstream is principled imputation of the half of the table we didn't get to see.

	customer	$Y(0)$ not emailed	$Y(1)$ emailed	individual effect	emailed?	\
0	Anna	30	80	50	yes	
1	Ben	55	60	5	no	
2	Cara	20	22	2	yes	

	$\rightarrow$ OBSERVED	$\rightarrow$ MISSING (counterfactual)
0	80	30
1	55	60
2	22	20

The trap in plain sight: comparing the **emailed** customers' *observed* spend to the **not-emailed** ones' is **not** the true average effect — that naive observed contrast is **confounded** by *who got chosen* to be emailed, and the gap between it and the real individual-effect average is the entire subject of this cookbook.

1.2 2 · From individuals to averages, and the enemy: confounding

Decisions run on **averages**: the **Average Treatment Effect** $ATE = \mathbb{E}[Y(1) - Y(0)]$. The whole difficulty collapses onto one word — **comparable**: to estimate $\mathbb{E}[Y(0)]$ for the treated we need a

control group *like them in every way except the treatment*.

Confounding breaks comparability. If we email our *loyal* customers and compare to *casual* controls, loyal customers spend more anyway — “loyalty” drove both who got emailed and how much they spend. A **confounder** is a common cause of treatment and outcome; the naive “treated — control” difference then blames the treatment for a gap that was really about *who ended up in each group*.

1.3 3 · The gold standard, and life without it

Randomization (an A/B test) annihilates confounding by design: a coin flip makes treatment independent of every confounder, measured or not, so $ATE = \mathbb{E}[Y | T=1] - \mathbb{E}[Y | T=0]$ — a plain difference of means. We can *see* this: the same simulator run in **randomized** vs **observational** mode gives the same true effect, but only the randomized naive estimate is unbiased.

The data-generating model behind this demo — `dgp.uplift_customers`, the notebook-01 simulator (its full equations are in notebook 01, step 2). Each customer has features x (recency R , frequency F , monetary M , tenure, engagement E), a known baseline $\mu_0(x)$, and a planted effect $\tau(x)$, with spend

$$Y = \mu_0(x) + \tau(x)T + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 8^2).$$

The **only** thing that differs between the two bars is how T is assigned:

randomized: $T \sim \text{Bernoulli}(\frac{1}{2})$ vs observational: $T \sim \text{Bernoulli}\left(\sigma(1.8(E-0.4)+1.2\frac{150-R}{150}+0.15(F-5))\right)$,

i.e. marketers historically emailed engaged, recently-active customers — the same characteristics that raise spend through $\mu_0(x)$. Identical true effect in both worlds; only the *assignment mechanism* changes, and that alone moves the naive estimate.

true ATE €5.6 · randomized naive €4.8 +/- 0.6 (SE) · observational naive €11.4 +/- 0.7 (SE)

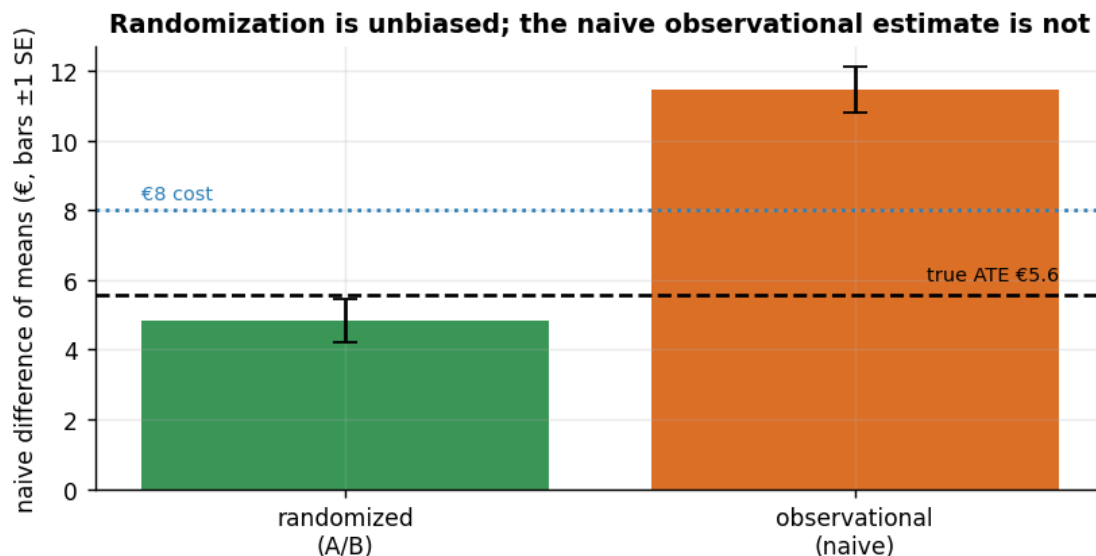
Same sampling noise (SE ~ €0.6 each), opposite reasons for missing the truth:

- randomized €4.8 is 1.2 SE from €5.6 — pure NOISE (unbiased; the gap shrinks like $1/\sqrt{n}$).

- observational €11.4 is 9.0 SE from €5.6 — that gap is BIAS (it does NOT shrink with n).

Decision at €8/contact: the TRUE effect €5.6 is BELOW €8, so emailing everyone does NOT pay.

But the confounded naive €11.4 is ABOVE €8 and would (wrongly) say 'email everyone' — the same identification error, now flipping a real go/no-go.



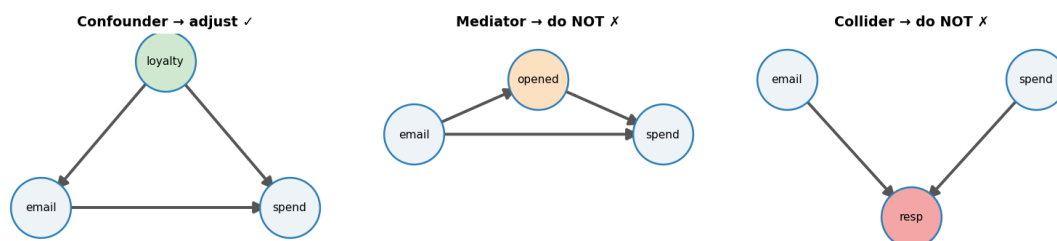
Without randomization we must **identify** the effect using assumptions:

1. **Unconfoundedness** $\{Y(0), Y(1)\} \perp T \mid X$ — within units alike on measured X , treatment is as-good-as-random. (*The big, untestable one.*)
2. **Positivity / overlap** $0 < e(x) < 1$, $e(x) = P(T=1 \mid X=x)$ the **propensity score** — every kind of unit could have gone either way.
3. **SUTVA** — no interference, one version of treatment.

Under these, the **adjustment (g-)formula** compares like-with-like *within* strata of X and averages back up: $ATE = E_X[E[Y \mid T=1, X] - E[Y \mid T=0, X]]$.

1.4 4 · Seeing vs doing (the do-operator), DAGs, colliders

Observing $X = x$ (units where X happened to be x , carrying all the reasons it got there) differs from **doing** $do(X = x)$ (reaching in and *setting* it). Causal effects are about *doing*. A **DAG** draws each variable as a node and each direct cause as an arrow, and tells you exactly what to control for:



- **Backdoor path** (arrow *into* T, e.g. $T \leftarrow \text{loyalty} \rightarrow Y$): block it by conditioning — the

minimal blocking set is the X in the adjustment formula.

- **Mediator** ($T \rightarrow M \rightarrow Y$): controlling for M removes part of the effect you want.
- **Collider** ($T \rightarrow C \leftarrow Y$): conditioning on C *manufactures* a spurious association. Notebook 05 is the whole “what to control for” story.

The third panel above draws exactly this: **response** is a *common effect* of email and spend (a collider), so conditioning on it opens a spurious email-spend path — which is why that panel is flagged “do NOT control.”

1.5 5 · One effect for everyone, or a different effect for each?

- **ATE** — one number (“did the program lift revenue overall?”): **program evaluation / incrementality** (notebooks 06–11).
- **CATE** $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X=x]$ — the effect as a function of *who*: **heterogeneous treatment effects** (notebooks 01–04). If τ were identical for everyone, there’d be no one to select — targeting works *because* $\tau(x)$ varies.

(Sometimes a method only recovers the effect for a sub-population — e.g. the compliers an instrument nudges. That restricted quantity is a **LATE**; notebook 11.)

1.6 6 · The estimator ladder — four ways to recover one effect

Where the machinery enters. On the **confounded** data (true effect €6), we climb a ladder of estimators, each leaning on different assumptions. On a **single** sample any estimate can land a little off — an honest 90% interval misses about 1 time in 10 — so the test of a *good* estimator is not “exact on this draw” but **unbiased**: centred on the truth *over repeated samples*, while the naive difference stays stubbornly biased. The right-hand panel below shows exactly that across many fresh samples.

1. **Naive** difference of means — ignores X , biased up.
2. **Regression adjustment** — model $\mathbb{E}[Y \mid T, X]$, average the contrast (right if the outcome model is right).
3. **IPW** — reweight by the inverse propensity so treated/control look alike (right if the propensity model is right).
4. **AIPW (doubly-robust)** — combine both; consistent if *either* model is right, reported here with a **bootstrap confidence interval** (AIPW is a frequentist estimator — this is *not* a Bayesian credible interval).

The data-generating model for the ladder — `dgp.dag_control_demo`, the same dataset notebook 05 dissects (equations there too). One **observed confounder**, loyalty $L \sim \mathcal{N}(0, 1)$, and an independent unobserved trait $G \sim \mathcal{N}(0, 1)$ that adds outcome noise but never touches T (so it is *not* a confounder):

$$T = \text{email} \sim \text{Bernoulli}(\sigma(0.8L)), \quad Y = \text{spend} = 20 + 5L + 6T + 3G + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 4^2).$$

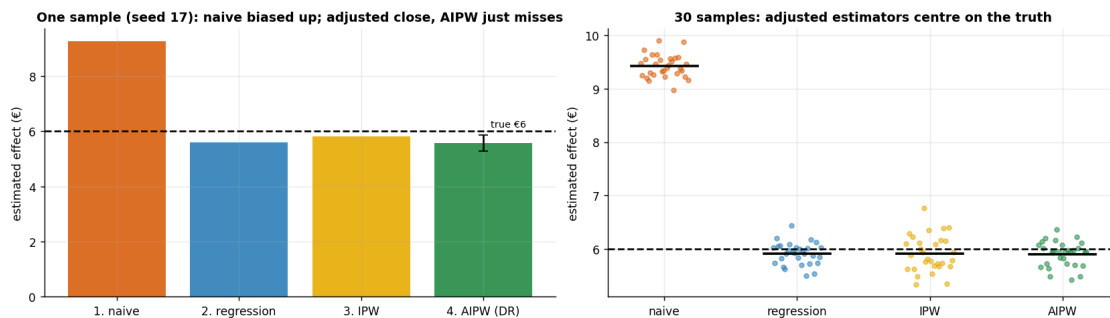
L raises both the emailing probability and spend — the open backdoor that rung 1 (naive) ignores and rungs 2–4 close, each in its own way: rung 2 models L in the outcome, rung 3 models L in the propensity, rung 4 does both. The true ATE the ladder must recover is the **6**.

1. naive	€ 9.26	(true €6.0)
2. regression	€ 5.60	(true €6.0)
3. IPW	€ 5.80	(true €6.0)
4. AIPW (DR)	€ 5.57	(true €6.0)

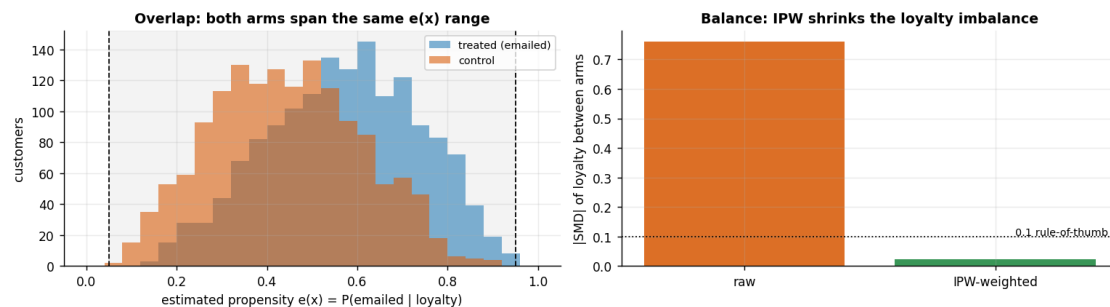
Across 30 fresh samples - mean estimate and bias vs the true €6:

naive	mean €9.42	bias +3.42	sd 0.21
regression	mean €5.91	bias -0.09	sd 0.20
IPW	mean €5.91	bias -0.09	sd 0.33
AIPW	mean €5.90	bias -0.10	sd 0.23

AIPW 90% interval covers the truth in 26/30 samples (87%); naive is the only estimator whose bias does not vanish.



overlap: e in $[0.06, 0.95]$; the clip binds on 0.0% of customers (rarely) - IPW shrinks the loyalty imbalance from SMD 0.76 to 0.02.



The discipline: **Bayes/ML sharpens estimation, never identification.** A tight, beautiful posterior computed under a false unconfoundedness assumption is *confidently wrong*. Priors don't fix broken causal arguments — the graph does. Every notebook keeps identification (this ladder's assumptions) separate from estimation (which rung you climb).

1.7 6b · The same lesson on real, famous data (LaLonde 1986)

The ladder above ran on our simulator. Here is the identical confounding story on the most famous real dataset in causal inference — **you don’t provide anything; it’s fetched from a public URL**:

LaLonde / NSW — a 1970s job-training program. Because it included a **randomized** arm, we know the *true* effect of training on 1978 earnings: about **+\$1,800**. But if we throw away the randomized controls and instead compare the trained workers to a *non-experimental* comparison group (people pulled from national surveys), a naive difference is **badly negative** — it looks like training *hurt* earnings, because the trained group was far more disadvantaged to begin with (the confounding). Adjusting for the measured covariates pulls the estimate back toward the true +\$1,800. This single dataset is why the field takes confounding seriously.

Gated on `CMP_REAL=1` (keeps the offline test suite deterministic).

Real-data section skipped. Set `CMP_REAL=1` and re-run to fetch the LaLonde dataset.

The payoff (what `CMP_REAL=1` prints). Fetching the real LaLonde/NSW data ($n = 614$; 185 trained) and running the *same two rungs* as our simulator gives:

estimator	effect on 1978 earnings	reading
naive difference of means	−\$635	training looks <i>harmful</i> — the comparison group started far richer
covariate-adjusted	+\$1,548	pulled back toward the ~ +\$1,800 experimental truth

The sign itself **reverses** once you adjust — the same confounding flip as the simulator, on the dataset that made the field take confounding seriously. (The committed offline run above shows only the skip message so the test suite stays deterministic; these numbers were produced by the gated cell with `CMP_REAL=1`.)

1.7.1 The eight words to walk in with

Treatment · **counterfactual** / **potential outcome** (the branch you didn’t observe) · **confounder** (common cause of treatment and outcome) · **randomization** (kills confounding by design) · **identification** (the assumptions that make an effect recoverable) · **do-operator** (forcing vs observing) · **backdoor** / **collider** (what to control for, and what never to) · **ATE** vs **CATE** (one effect for all vs a different effect per customer).

Next: 01 · [uplift targeting](#) (Anchor A, HTE) and 07 · [geo-lift synthetic control](#) (Anchor B, program evaluation) — then the rest of the cookbook.